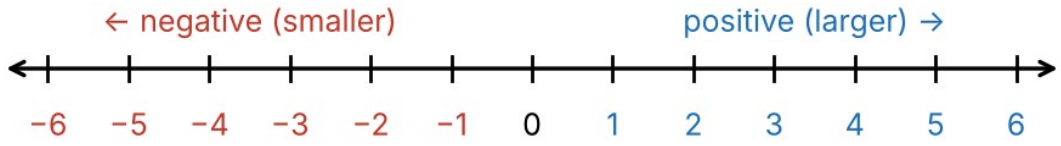
Help Sheet

Keep this beside you while you do the practice exams. Try each question first, then use this sheet if you get stuck. No calculators – all methods here are by hand.

1. Number

Part A Q2–Q5, Q11 · Part B Q5, Q10

Topic	What you need to know	Example
Order of operations	Brackets first then × and ÷ (working left to right) then + and − (working left to right)	5 + 2 × (7 − 3) = 5 + 2 × 4 = 5 + 8 = 13
Ascending and descending	Ascending = smallest to largest (going UP the stairs) Descending = largest to smallest (going DOWN)	Ascending: −3, 0, 2.5, 8 Descending: 8, 2.5, 0, −3
Place value of decimals	After the decimal point: 1st place = tenths 2nd place = hundredths 3rd place = thousandths 4th place = ten thousandths	In 4.2617: 2 is tenths, 6 is hundredths, 1 is thousandths, 7 is ten thousandths
Index notation	The index (power) tells you how many times to multiply the base by itself. It does NOT mean base × index.	$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$ $4^3 = 4 \times 4 \times 4 = 64$ (not 12!)
Negative numbers (integers)	Numbers left of zero are negative. The further left, the smaller. Adding a positive → move right. Subtracting a positive → move left.	−4 + 9 = 5 −2 − 6 = −8 −7 + 3 = −4 3 × (−2 + 7) = 3 × 5 = 15
Mental strategies	Look for pairs that make 10, 20, 50 … × by hundreds: multiply, then add the zeros. Round and adjust, or split the number.	16 + 9 + 4 + 11 = 20 + 20 = 40 3 × 400 = 12 hundreds = 1200 18 × 4 = 20 × 4 − 2 × 4 = 72 23 × 6 = 120 + 18 = 138



2. Fractions and decimals

Part B Q3, Q4, Q6

Topic	What you need to know	Example
Equivalent fractions	Multiply (or divide) the numerator AND the denominator by the same number. If the top and bottom are not multiplied by the same number, the fractions are NOT equal.	$\frac{2}{3} = \frac{8}{12}$ True (× 4) $\frac{3}{5} = \frac{9}{20}$ False (3 × 3 = 9 but 5 × 4 = 20)
Multiplying fractions	Multiply the tops, multiply the bottoms, then simplify.	$\frac{3}{4} \times \frac{2}{5} = \frac{6}{20} = \frac{3}{10}$
Multiplying a decimal	Ignore the decimal point, multiply, then put back the same number of decimal places.	0.7 × 4: 7 × 4 = 28 → 2.8 0.9 × 3: 9 × 3 = 27 → 2.7

2. Fractions and decimals

Part B Q3, Q4, Q6

Topic	What you need to know	Example
Improper fraction → mixed number	Divide the numerator by the denominator. The answer is the whole number; the remainder goes over the denominator.	$\frac{11}{3}$ → 11 ÷ 3 = 3 r 2 so $\frac{11}{3} = 3\frac{2}{3}$
Mixed number → improper fraction	Whole number × denominator, + numerator. Keep the same denominator.	$2\frac{3}{4}$ → 2 × 4 + 3 = 11 so $2\frac{3}{4} = \frac{11}{4}$

3. Algebra

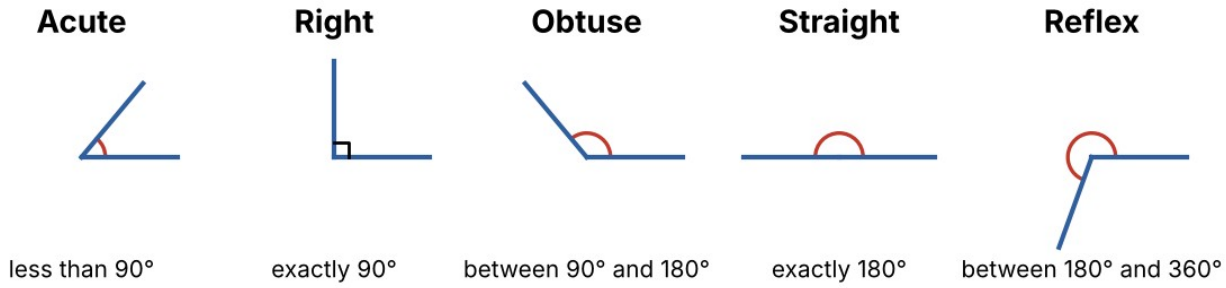
Part A Q7 · Part B Q7, Q8, Q9, Q11

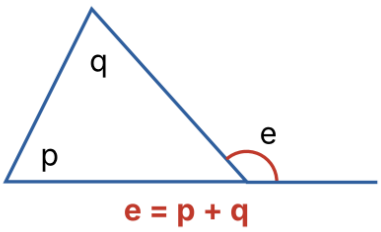
Topic	What you need to know	Example
Variable (pronumeral)	A letter that stands for a number we don't know yet.	In 3n + 2, the letter n is the variable.
Writing expressions	more than, sum, increased by → + less than, difference, decreased by → − times, product, double → × divided by, quotient, half → ÷ No equals sign in an expression.	'4 more than a number' → x + 4 '4 less than a number' → x − 4 'the product of 3 and a number' → 3x 'half a number' → $\frac{x}{2}$
Substitution	Replace each letter with its number, then follow the order of operations. Letters next to each other are multiplied: ab means a × b.	If a = 3 and b = 5: 2a + b = 2 × 3 + 5 = 11 ab − 1 = 3 × 5 − 1 = 14
Collecting like terms	Like terms have exactly the same letters. Add or subtract the numbers in front. Numbers on their own stay separate.	3y + 5y = 8y 4ab + ab = 5ab (ab means 1ab) 6p + 3 − 2p = 4p + 3
Multiplying terms	Multiply the numbers, then write the letters.	4 × 5t = 20t 2 × 3y = 6y
Solving one-step equations	Do the inverse (opposite) operation to both sides: + ↔ − × ↔ ÷ Check by substituting your answer back in.	a + 5 = 12 → a = 12 − 5 = 7 b − 4 = 10 → b = 10 + 4 = 14 3c = 21 → c = 21 ÷ 3 = 7 $\frac{d}{2}$ = 9 → d = 9 × 2 = 18
Solving two-step equations	Undo + or − first, then undo × or ÷. If the whole top of a fraction is divided, multiply both sides first.	4a + 3 = 23 −3: 4a = 20 ÷4: a = 5 $\frac{k+1}{3}$ = 5 ×3: k + 1 = 15 −1: k = 14

4. Angles and shapes

Part A Q8–Q10, Q14, Q15 · Part B Q1, Q2

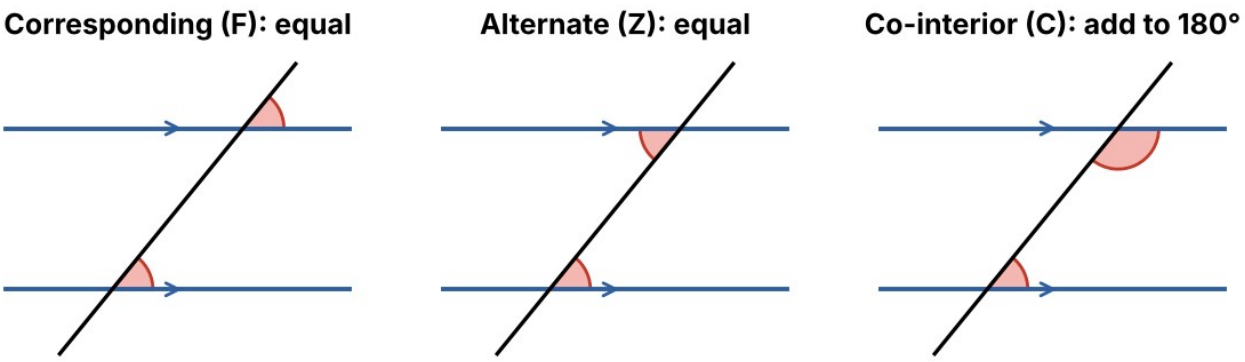
Topic	What you need to know	Example
Angle sum of a triangle	The three angles inside a triangle add to 180°. Missing angle = 180° − (the other two). Write the reason: 'angle sum of a triangle is 180°'.	Angles 65°, 40° and x: x = 180° − 65° − 40° x = 75°
Types of triangles	Isosceles: 2 equal sides and 2 equal angles (opposite the equal sides). Equilateral: 3 equal sides, all angles 60°. Scalene: no equal sides.	Isosceles with equal angles of 50°: third angle = 180° − 50° − 50° = 80°
Exterior angle of a triangle	Extend one side. The exterior angle = the sum of the two opposite interior angles. (Or: 180° − the interior angle next to it, because angles on a straight line add to 180°.)	Opposite interior angles 45° and 70°: exterior angle = 45° + 70° = 115°
Perimeter	Perimeter is the total distance around the OUTSIDE of a shape. Add all the side lengths. (Area is the space inside – don't mix them up.)	Rectangle 7 cm by 2 cm: P = 7 + 2 + 7 + 2 = 18 cm





Angles on parallel lines

Corresponding angles (F shape) are equal.
Alternate angles (Z shape) are equal.
Co-interior angles (C shape) add to 180° (supplementary).
Supplementary = add to 180°. Complementary = add to 90°.



5. Statistics (data)

Part A Q1, Q2 · Part C Q12, Q13, Q14

Topic	What you need to know	Example
Mean	Add all the values, then divide by how many values there are.	5, 8, 3, 8, 6 Mean = 30 ÷ 5 = 6
Median	Put the values in order. The median is the middle value. If there are two middle values, add them and halve.	In order: 3, 5, 6, 8, 8 → median = 6 2, 4, 7, 9 → (4 + 7) ÷ 2 = 5.5
Mode	The value that occurs most often. (MOde = MOst)	5, 8, 3, 8, 6 → mode = 8
Range	Highest value − lowest value.	5, 8, 3, 8, 6 → 8 − 3 = 5
Adding a new value	Work everything out again with the new value included. New mean = (old total + new value) ÷ (new number of values). Re-order the data for the new median.	Add 12 to 5, 8, 3, 8, 6: new mean = (30 + 12) ÷ 6 = 7 new median: 3, 5, 6, 8, 8, 12 → (6 + 8) ÷ 2 = 7
Outlier	A value that is much higher or much lower than the rest of the data.	2, 3, 3, 4, 3, 15 → outlier is 15

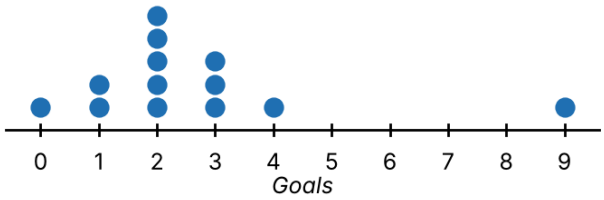
Constructing a dot plot

1. Rule a horizontal number line.
2. Mark every value from the lowest to the highest, evenly spaced – even ones with no dots.
3. Put one dot above the value for each piece of data. Keep dots evenly stacked.
4. Label the axis and give it a title.

Total = count all the dots.

'Less than 3' means 0, 1 and 2 – NOT 3.

Example: goals scored by a netball team



Column graphs	Read across from the top of each bar to the scale. Total = add the height of every bar. A day has 24 hours, so a fraction of a day = hours ÷ 24. Simplify!	8 hours of a day: $\frac{8}{24} = \frac{1}{3}$ 3 hours of a day: $\frac{3}{24} = \frac{1}{8}$

6. Probability

Part A Q12, Q13 · Part C Q15

Topic	What you need to know	Example
Sample space	A list of every possible outcome, each written once, inside curly brackets { }.	Die: {1, 2, 3, 4, 5, 6} Coin: {heads, tails} Spinner: {red, blue, green}
P(E)	P(E) means 'the probability of event E happening'. P(E) = number of favourable outcomes ÷ total number of outcomes. On a spinner with equal sections, count the sections.	Spinner with 5 equal sections, 2 are red: P(red) = $\frac{2}{5}$ Die: P(even) = $\frac{3}{6} = \frac{1}{2}$

6. Probability

Part A Q12, Q13 · Part C Q15

Topic	What you need to know	Example
Probability scale	Probabilities are between 0 and 1. 0 = impossible $\frac{1}{2}$ = even chance 1 = certain	P(rolling a 7 on a die) = 0 P(rolling less than 7) = 1
Complement	The complement of E is ‘E does NOT happen’ – all the other outcomes. P(not E) = 1 − P(E).	P(red) = $\frac{2}{5}$ Complement: not red P(not red) = 1 − $\frac{2}{5} = \frac{3}{5}$

Exam tips

- About one minute per mark. If you are stuck, move on and come back.
- Multiple choice: read all four options, cross out the ones that are wrong, then circle ONE letter.
- Show your working – marks are given for the method, not just the answer.
- Check equations by substituting your answer back in. Check probabilities are between 0 and 1.
- Bring a black pen, pencil, ruler and eraser. A calculator is NOT allowed.